General Science

Alveary

About the Course

This is the first of a two-year Physical Science course (i.e., introductory physics and chemistry). The fullness of the course includes topics in earth science, history, politics, and more. The completion of this course provides a gentle yet robust transition into the high school disciplines. The lab component is required for this course. Coordinating afternoon activities are provided in Outdoor Work.

This topic is included in the following course(s): Science: Grade 7

About Science: Grade 7
In level 7, learners extend their relationship with science in a new direction (time), situating science in its historical, political, and cultural context, and beginning to understand that science is an ever-changing process rather than a static body of knowledge. Reading level and expectation increase from previous levels.

Science: Grade 7
The Big Picture:
To accomplish the goal of supporting a relationship with the Things of the Universe, a Mason science program consists of nature lore, natural history, and general science. Nature immersion, inquiry, community connection, and supportive literature are woven into each of these three parts. In Form 1, students became acquainted with this area of knowledge, science. Form 2 students extended their scope and began to encounter science as an active part of society. Form 3 students extend their relationship with science in a new direction (time) and consider science as a process by which man seeks Truth within Creation. They begin to situate science in its historical, political, and cultural context with all of the complexities of man's own story. In this way, students are prepared to think about science with a complete and holistic perspective.

Nature lore is timeless knowledge that is passed through a community, much like a grandmother passes on how to make that special bread when the dough just "feels right." Like Mason, we strive to pass on this knowledge primarily through outdoor work. Group nature walks, seasonal readings, and topics in scouting are provided as an Outdoor Work resource in the Quick Links. If desired, literature suggestions to support lore can be found in the Community Read Alouds resource (in Citizenship Grades 4+).

Placement & Combining Tips

About Science: Grade 7
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General Science: Grade 7
For seventh-grade students or possibly hungry sixth-graders or eighth-graders taking their time. If combining grades is desirable: the combined grades may begin at the appropriate level and move forward together, or teachers can adapt the reading level and laboratory content to meet their needs.

Scheduling

GRADE	SCHEDULE INFO.	BOOKS
7	General Science: Grade 7 2 times/week 30 min	The Story of Science: Aristotle Leads the Way

Sample Weekly View

Day 1	Day 2	Day 3	Day 4	Day 5
Science: Grade 7				
Natural History: Grade 7	General Science: Grade 7	General Science: Grade 7	Nature Notebook: Grade 7	Labs: Grade 7 Nature Walks & Scouting: Grades 1-8

Planning & Prep

Permission to print for non-commercial use. See Alveary group use policy to use lessons in a group context.

LINKS: Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.

Science: Grade 7
☐ Obtain any supplies indicated on the science or grade-level supply lists.
☐ Download any apps and shortcut any desired links.

Nature Lore:
☐ Bookmark your Outdoor Work Quick Link, so that you have it available on your weekly outing. Outdoor Work is generally flexible for your location and season and can be moved around in the schedule to incorporate or substitute Natural History Club outings, except where the suggestion is to collect some Thing for a lesson.
☐ Print or bookmark grade-specific nature notebook suggestions to support natural history and general science. Notebooking can be done on a walk, during occupations, or as a field trip, as appropriate.

Special Topics & Field Trips

Science: Grade 7
Encourage students to choose their own special topic and to notice its ecological relationships. What or who are they curious about or interested in getting to know better? Teachers can choose their own special study, too!

☐ Learn a few of your local species or varieties in connection with special topics.

General Science: Grade 7
☐ Terms 1 and 3: a planetarium or local astronomy club event
☐ Term 2: a playground

Term Prep & Teacher Tips

General Science: Grade 7
☐ Gather household items, typically easy for students to scavenge or teachers to obtain locally:
• medium-large sized recycled box, roughly 1-2'
• knife, blade, or scissors to cut the box
• table/reading lamp
• table or countertop
• books or wood blocks to adjust height
• 1 gallon jug of milk or water, the type with a handle
• horizontal bar to support weight, such as broom handle held by 2 people. a chin up bar, a porch railing, etc.

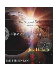
- recycled cereal box or toothbrush
- a flat location that is in the sun at noon
- various supplies by student design
- printables
- computer
- optional: empty cereal box
- optional: orange
- optional: flashlight
- optional: calculator, such as found on most electronic devices

Books & Resources

For book rationales and purchase options, click the Book List link or scan the QR code below.

∞ View Book List Details

General Science: Grade 7



The Story of Science: Aristotle Leads the Way

Supplies

For supply list details and basic supplies helpful to have on hand, click the links or scan the QR code below.

∞ View Basic Supplies
∞ View Supply List Details

General Science: Grade 7

Household Items - General Science: Grade 7

Quick Links

Click THIS text or scan the QR code for links.

Related Course Quick Links

∞ Extra Helpings
∞ Foundations (See Section 13: Science)
∞ Alternate One Year Physical Science Lesson Plans
∞ Outdoor Work
∞ Seek app from iNaturalist
∞ SkyView Lite for Android
∞ SkyView Lite for iOS

General Science: Grade 7

How To Teach

Introduce

• Begin each lesson with learners' existing knowledge. If the book or activity is new or unfamiliar, then look at the title, a picture, or guidance in the lesson plan to help discuss what students think, drawing on previous experience. If continuing or revisiting a topic or activity, then recap.

• Often, the introduction in the lesson plans or the title of the book's section can help learners to draw out the main idea. Use this to support learners, as appropriate.

• Some learners may benefit from using pictures or looking back briefly.

• Allow them time to share any concerns and come alongside, as needed.

Read

• Read or do, as instructed in the lessons, noting any Teacher Tips provided. Learners should always have their journal, notebook, or other paper available in case they need to draw or diagram during the lesson.

• Use supportive strategies and educational tools to reduce frustration and better engage the mind, as appropriate. These could include, but are not limited to, the use of eBooks, pictures, audio, read-aloud, buddy reading, colored reading strips, etc.

• If learners do not understand a word or concept, do not worry. Try to show them with a picture or connect the idea to something they have seen in real life. They are learning much by the way and will likely build understanding over the term, the year, and beyond.

Narrate

• Process the ideas of the lesson by retelling events, describing, explaining a concept, etc. Tips about helping learners deal with various types of passages are provided, but teachers can learn more in Charlotte Mason's School Education Chapter 16, especially p.180.

• Learners may use words, pictures, Legos, PowerPoint, etc, to process and convey ideas in their own way.

• Teachers may take turns to model.

Discuss

• Consider together any thoughts, confusion, or concerns about the passage, keeping in mind that oftentimes new concepts are not going to be 'mastered' on the first or even second introduction. Mastery is not necessarily the goal, but curiosity and thinking.

• Questions/topics for further discussion are often provided in the lesson plans (or even lab books) to help. There are no right or wrong answers to these. Alternatively, many of these can be used for composition, depending on the needs of the learner and the instructional goals of the teacher. If teachers want to keep ideas active in the mind, these can also be used at other times to keep the ideas in the working memory.

• Note that learners may need to spend more than the allotted time engaging with or even repeating a lesson before moving on or as reinforcement at a later time. Adjust the pace as needed to feed the learner.

• Notice if there were any dates that they want to keep for their Book of Centuries.

Connect

• Follow any extra links, examine any sidebars in the text, look at pictures, etc., depending on learner needs and interest.

• These can be viewed as alternative ways to engage.

General Science: Grade 7

[Click THIS text or scan the QR code for links.]

Term 1

WEEK 1

⬚ **30m General Science: Grade 7 - Lesson 1**
Thinking About Creation

▯ Materials: Aristotle Leads the Way

PREP: Read Teacher Tip

→ INTRO
This book tells the story of how modern science developed, beginning with the ideas of the ancient Greeks. Most of them wondered and learned from the heavens and from mathematics, eventually defining science. There are many interesting sidebars and tangents in the text, but read the main text first and then explore them, as desired. Watch the course introduction video:
∞ Video Link: Form 3 Science Intro (5:04)

→ READ, NARRATE, & DISCUSS
Ch.1 p.1-8 "The universe" - "to have them."

• Tell about the earliest beginnings of science in the ancient world. How did men begin?
• The author of this book invites all learners of different beliefs to find common ground in this discussion about how we understand the world we share. Discuss some of the ways she does this.

→ SUPPLEMENTAL
∞ Video Link: NatGeo Mesopotamia (4:10)
∞ Video Link: Mayan Creation Story (2:55)
∞ Video Link: Hindu Creation Story (2:41)

★ TEACHER TIP
Watch the course introduction video with your student(s).

★ TEACHER TIP
Even as students begin to work more independently, teacher engagement is very important. Discussion prompts are provided to help, if natural discussion is a challenge.

★ TEACHER TIP
Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, to enliven the lesson itself, or to add to discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s).

• IMPORTANT DATES
Sumerian Empire (c.3100 B.C.-c.2000 B.C.), Egyptian Civilization (c.3100 B.C.-332 B.C.)

WEEK 1

⬚ **30m General Science: Grade 7 - Lesson 2**
What is a Myth?

▯ Materials: Aristotle Leads the Way

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS
Ch.2 p.9-14 "Once, or so" - "makes us human."

• According to the author, what is the distinction between myth and science? And what is NOT the distinction? Is there anything to add based on your own tradition?
• Why is myth important? If you have read much of J.R.R.Tolkien's work, consider the role myth plays in his stories. Compare/contrast both authors' ideas about myth. If you haven't yet read his work, consider another author who makes use of myth, or file away this idea for the day that you do read his work!
• What is a hypothesis?
• How is science (and its limits) part of what makes us human?

→ SUPPLEMENTAL
∞ Video Link: Finding Summer Constellations (5:39)

★ TEACHER NOTE
The author invites readers to consider what is myth and what is science, a consideration that engages with religion, on p.11-12: "How do we" - "fiction usually does." Teachers might choose to allow the idea to pass by the way or to engage in discussion.

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[Click THIS text or scan the QR code for links.]

Term 1

WEEK 2	⬚ **30m General Science: Grade 7 - Lesson 3** *What is a Myth?* Materials: Aristotle Leads the Way **→ RECAP** What did you read last time? **→ READ, NARRATE, & DISCUSS** Ch.2 p.14-19 "We've learned" - "to be born." • Describe the process of science. • "The world is God's epistle written to mankind... It was written in mathematical letters." Discuss. • The author says that the "marriage of numbers and nature... is an essential part of the scientific story." Why? Do you agree? Where have you noticed numbers in nature? • Why do you suppose the story begins with sky watching? **→ SUPPLEMENTAL** ∞ Video Link: Is Math Invented or Discovered? (5:11)	**• NATURE NOTEBOOK** Prompt for General Science in Outdoor Work Quick Link.
WEEK 2	⬚ **30m General Science: Grade 7 - Lesson 4** *Thinking about Time* Materials: Aristotle Leads the Way **→ RECAP** What did you read last time? **→ READ, NARRATE, & DISCUSS** Ch.3 p.20-25 "Perhaps it was" - "around the globe." • What kinds of observations and what kinds of questions did early sky watchers have? • Explain the similarities and differences between the lunar calendar and the solar calendar. Which is closer to the calendar we use? **→ SUPPLEMENTAL** ∞ Video Link: The Analemma (4:23) ∞ Video Link: Want to measure the analemma? (7:33)	**• IMPORTANT DATES** Egyptian Solar Calendar (c.4th century B.C.)
WEEK 3	⬚ **30m General Science: Grade 7 - Lesson 5** *Phases of the Moon* Materials: Aristotle Leads the Way **→ RECAP** What did you read last time? **→ READ, NARRATE, & DISCUSS** Ch.3 p.26-27 "For a long time," - "facing the sun." • Explain the moon's phases. **→ SUPPLEMENTAL** ∞ Video Link: Moon Phases (9:45)	**• NATURE NOTEBOOK** Prompt for General Science in Outdoor Work Quick Link.

General Science: Grade 7

[Click THIS text or scan the QR code for links.](#)

Term 1

WEEK 3

⬚ **30m General Science: Grade 7 - Lesson 6**
Thinking about Time

▯ Materials: Aristotle Leads the Way

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS
Ch.3 p.28-31 "Palenque," - "our solar system."

• Were the ancient calendar makers doing science? Why or why not? If not science, what was it?
• What ideas have been passed down to us from the ancient calendar makers?

→ SUPPLEMENTAL
∞ Video Link: Decoding the Astronomy of Stonehenge (6:29)

• **FIELD TRIP**
Visit a planetarium or astronomy club event.